

- Q.24 Explain the IUPAC naming of cycle and aromatic compounds with two examples each.
- Q.25 Describe the manufacturing process of methanol from syngas with flow sheet.
- Q.26 Write down five major uses of formaldehyde?
- Q.27 List and briefly explain five functional groups.
- Q.28 Give the reaction and flow sheet for ethylene oxide manufacturing by direct oxidation of ethylene.
- Q.29 Write five properties of ethanol.
- Q.30 Describe the manufacturing process of acetone by the cumene hydroperoxide process. Give reactions involved.
- Q.31 Describe the raw materials used in the petrochemical industry.
- Q.32 Write down the properties and uses of butadiene.
- Q.33 Describe the manufacturing of Styrene Butadiene Rubber (SBR) with flowsheet.
- Q.34 Describe the manufacturing process of styrene from ethylbenzene with major engineering problems
- Q.35 Write a note on the manufacture of phthalic anhydride from naphthalene. Give the flowsheet and properties.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What are the C_3 compounds? Explain the cumene production process with flowsheet and write down the necessary reactions involved.
- Q.37 Describe in detail the Manufacturing of Vinyl Chloride with flowsheet. Discuss its properties and uses.
- Q.38 Write a comprehensive note on Aromatic Compounds. Explain the manufacturing process of phenol. raw a neat flowsheet and mention related major engineering problems.

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Roll No.

5th Sem / Chemical Sub : Petrochemicals Technology

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is the primary raw material for the petrochemical industry?
- Coal tar
 - Naphtha / Natural gas
 - Vegetable oils
 - Biomass
- Q.2 The process of thermally cracking raphtha to produce Olefins (ethylene, propylene) is called:
- Catalytic reforming
 - Naphtha cracking (steam cracking)
 - Hydrocracking
 - Visbreaking
- Q.3 The IUPAC name of CH_2-CH_2-OH is:
- Methanol
 - Ethanol
 - Propanol
 - Butanol
- Q.4 The function group - CHO is characteristic of:
- Ketones
 - Carboxylic acids
 - Aldehydes
 - Esters
- Q.5 Methanol is industrial manufactured by:
- Fermentation of sugars
 - Catalytic hydrogenation of CO and CO_2 (syngas process)

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- c) Oxidation of ethane
 - d) Hydration of ethylene
- Q.6 Vinyl chloride monomer is used as the raw material for making:
- a) Polyethylene
 - b) PVC
 - c) Polypropylene
 - d) Nylon
- Q.7 Ethylene oxide is manufactured by:
- a) Dehydration of ethanol
 - b) Catalytic oxidation of ethylene
 - c) Chlorination of ethylene
 - d) Hydration of acetylene
- Q.8 Cumene is used as an intermediate to manufacture which two important chemicals?
- a) Acetone and methanol
 - b) Phenol and acetone
 - c) Acetic acid and acetaldehyde
 - d) Acrylonitrile and butadiene
- Q.9 SBR is a copolymer of:
- a) Styrene and propylene
 - b) Butadiene and acrylonitrile
 - c) Styrene and butadiene
 - d) Butanol and styrene
- Q.10 Phthalic anhydride is manufactured by:
- a) Oxidation of naphthalene or O-Xylene
 - b) Hydrogenation of benzene
 - c) Chlorination of toluene
 - d) Dehydration of phthalic acid in cold

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The two major petrochemicals obtained directly from naphtha cracking are _____ and _____.
- Q.12 Write down the IUPAC name of CH_3COCH_3 .
- Q.13 Cumene is a C_2 compound (True/False)
- Q.14 List any one major petrochemical plant in India.
- Q.15 Name the catalyst used in methanol manufacturing from synthesis gas.
- Q.16 Write down the functional group used for esters.
- Q.17 The IUPAC suffix for a carboxylic acid functional group is "-ol". (True / False)
- Q.18 Define petrochemicals.
- Q.19 The manufacturing process of acrylonitrile by oxidation of propylene with ammonia and oxygen is called the _____ process.
- Q.20 Maleic anhydride is used in the manufacture of unsaturated polyester resins. (True / False)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List five major petrochemicals derived from petroleum products and their uses.
- Q.22 Write a note on naphtha cracking. What are the major products obtained?
- Q.23 Write the IUPAC names of the following
- i) $\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$
 - ii) CH_3CHO
 - iii) CH_3COOH
 - iv) $\text{CH}_3\text{COOCH}_2\text{CH}_3$
 - v) $(\text{CH}_3)_2\text{CO}$